CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

AI engineer (MSc in Artificial Intelligence, 3 years of experience) who prepares data, runs AI models end to end and owns their output quality. I turn business decision workflows into working AI solutions with stakeholders, validate model output against ground truth (precision/recall 33%/45% → 93%/90% at Kniit) and improve it through fine-tuning and optimization cycles. Python, SQL, scikit-learn, PyTorch, AWS and Azure.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning an AI system end-to-end for the German energy sector: data ingestion, structuring and chunking of regulatory and technical documents, prompt engineering, and indexing into a knowledge-graph data model, working with SAP domain experts and reporting to the CTO.
- Designed and optimized multi-agent LangGraph extraction (entity-extraction, relationship-analysis, consolidation, and reflection agents), tracking quality metrics against an SAP-system-derived ground truth over repeated optimization cycles and raising precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) pipeline for a second, high-volume data stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Developed Expert Interface, a web app (React, Python microservices) on Azure where domain experts review and approve changes to the data model.
- **Tech stack: Python, SQL (Postgres), Neo4j, MongoDB, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised on how agentic AI could support them, and translated their needs into a solution approach.
- Designed and deployed an agentic AI application with LangGraph that optimizes these decision-making workflows, routing each request to vector database queries, general inference, or summarization.
- Built the full stack (React frontend, Python/FastAPI backend), containerized it with Docker, deployed it to AWS with CDK, and version-controlled it in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated validation of AI output quality (RAG generation and retrieval) with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data and fine-tuned models for optimized retrieval.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- **Tech stack: Python, PyTorch, Hugging Face Transformers, LangChain, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Prepared and structured sensor time-series data (lag features, rolling statistical windows) and built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity).
- Benchmarked the models against empirical curve-fitting and improved early identification of batches likely to deviate from specification.
- Built a Streamlit dashboard letting chemists simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Pandas, NumPy, scikit-learn, XGBoost, TensorFlow/Keras, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.

PROJECTS

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA
- Fine-tuned an Open Llama 3B model with QLoRA (for faster, lower-memory training) to generate responses to educational meme-related prompts.

Project: Knowledge Graph | NLP (NLTK) | Protégé | SPARQL | RDF/OWL
- Modelled a knowledge graph as a learning resource for AI students with Protégé and rdflib: information extraction with NLTK, RDF transformation, SPARQL querying, OWL enhancement.

TECHNICAL SKILLSET

- **Data & Modelling:** Python, SQL (Postgres), Pandas, NumPy, data structures and data models (Neo4j knowledge graphs, RDF/OWL, SPARQL), MongoDB, DynamoDB, vector databases
- **AI Models:** scikit-learn, PyTorch, TensorFlow, XGBoost, Hugging Face Transformers, model fine-tuning (QLoRA), time-series forecasting
- **GenAI & Agentic AI:** LangGraph, LangChain, LLM APIs (LiteLLM), RAG, GraphRAG, prompt engineering, output validation (RAGAS, PytrecEval), MCP, Claude Code
- **Cloud & Tools:** AWS (CDK, Lambda, EC2, SQS, S3), Azure, Docker, Kubernetes, FastAPI, Streamlit, Git, GitLab CI/CD

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering (July 2019 – July 2021)
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT
- Udemy - The Complete Neural Networks Boot camp
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)